USN | | | | | | | | | | | [22CE513]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER -2024

RAILWAY, AIRPORT AND HARBOR ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
			BL	**CO**	**M**
1.	**a.**	How does coning of wheels influence the stability and performance of trains?	3	01	**02**
	b.	Provide a brief overview of gauge widening in railway tracks.	2	02	**02**
	c.	Examine how the International Civil Aviation Organization (ICAO) classifies airports.	3	03	**02**
	d.	Evaluate two design interventions that improve circulation in an airport.	5	04	**02**
	e.	Discuss the importance of dolphins in harbors.	2	05	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Write short notes on rails.	2	01	**05**
		OR			
	b.	Discuss the importance of rail transport in today's transportation system.	2	01	**05**
3.	**a.**	Provide a detailed analysis of the construction process for railway tracks.	4	06	**05**
		OR			
	b.	Calcuate the quantity of materials required per kilometer length of broad gauge (BG) track for a rail length of 12.8 m. Given that the sleeper density is equal to M+4 and the rail type is 90R with a weight of 44.7 kg/m, provide a detailed calculation.	4	02	**05**
4.	**a.**	Analyze the advantages of air transport.	4	03	**05**
		OR			
	b.	Analyze the impact of various parameters on the geometric design of runway.	4	03	**05**
5.	**a.**	How does the Air Traffic Control Network function to ensure safe and efficient air traffic management?	3	04	**05**
		OR			
	b.	Give a detailed analysis of the significance and details of runway markings.	3	04	**05**

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6.	**a.**	Analyze the key functions and importance of wharves in maritime transportation.	4	05	**05**
		OR			
	b.	Evaluate the role of inland water transportation in enhancing trade and connectivity in regional economies.	4	05	**05**
7.	**a.**	Describe the basic components of a helipad and explain the function of each.	2	04	**05**
		OR			
	b.	Summarize the key factors that should be considered in the design, layout, and construction of a railway yard.	2	02	**05**
		PART-C		**6 X 10=60**	
8.	**a.**	Examine the components of a permanent way, examining their roles and interactions within the railway system, and provide a neat sketch to illustrate them.	4	01	**10**
		OR			
	b.	Examine the phenomenon of creep in rails, including its control measures and the potential consequences it can have on railway infrastructure.	4	01	**10**
9.	**a.**	Analyze the various defects that can occur in rails and discuss their potential impacts on railway safety and performance.	4	02	**10**
		OR			
	b.	Examine the critical factors to consider in the design, layout, and construction of a railway station.	4	02	**10**
10.	**a.**	Evaluate how different aircraft characteristics influence the design and planning of airports.	4	03	**10**
		OR			
	b.	Evaluate the wind data collected over the last 10 years for a city and determine the optimal runway orientation that ensures 100% wind coverage.	5	06	**10**

Wind direction	**Duration of wind (%)**		
	6.4-25 kmph	**25-40 kmph**	**40-60 kmph**
N	0.4	1.7	0.4
NNE	2.1	5.8	0.9
NE	6.2	7.6	0.8
ENE	4.2	4.2	1.1

E	0.6	0.6	0.2		
ESE	2.1	0.4	0.1		
SE	1.4	0.5	0.2		
SSE	1.6	1.8	0.1		
S	1.6	1.2	0.1		
SSW	3.7	5.8	0.3		
SW	4.8	6.2	0.9		
WSW	4.6	4.8	0.4		
W	1.7	1.4	0.1		
WNW	1.4	1.1	0.2		
NW	1.8	0.8	0.1		
NNW	1.5	1.2	0.2		

11.	**a.**	Explore the various types of motor vehicle parking areas at an airport, providing a detailed discussion of each type.	3	04	**10**
		OR			
	b.	Provide a detailed analysis of various visual aids used in airport & its significance.	3	04	**10**
12.	**a.**	What factors need to be taken into account when designing a port, and what defines the characteristics of a good port?	3	05	**10**
		OR			
	b.	Provide a detailed overview of the components of a harbor, including the functions of each structure and their specific locations within the harbor area.	3	05	**10**
13.	**a.**	Analyze the implications of the Coastal Regulation Zone (CRZ) Notification of 2011 on coastal development and environmental protection.	4	05	**10**
		OR			
	b.	Assess the significance of road and water transport. Evaluate how the coordination of these modes can contribute to achieving sustainability in transportation systems.	4	06	**10**